A Novel Super-Resolution Model for 10-m Mangrove Mapping With Landsat-5

Wei Chen, Jinyan Tian, Jie Song, Xiaojuan Li, Yinghai Ke, Lin Zhu, Yongxin Yu, Yang Ou, and Huili Gong

***Abstract*— Existing temporal mangrove products are at a 30-m resolution from Landsat, facing challenges such as unclear delineation of mangrove community edges, difficulty in identifying creeks and open spaces within communities, and ineffective recognition of small patches. Therefore, there is an urgent need to produce higher resolution temporal mangrove products (e.g., 10-m) with Landsat, particularly considering the absence of available Sentinel imagery before 2015. To this end, we propose a novel super-resolution model that incorporates residual channel attention networks (RCAN) and texture transformer network (TTSR) to generate 10-m Landsat-5, namely RCAN-TTSR. RCAN and TTSR play crucial roles from different perspectives in the super-resolution process, respectively. TTSR accurately transfers texture information from Sentinel-2 to Landsat by computing the texture correlation between them. On the other hand, RCAN assigns different weights to multiple low-frequency features and a small number of high-frequency features derived from the raw bands of Landsat imagery, thus achieving better super-resolution outcomes. The results demonstrate that images produced by this model significantly outperform existing super-resolution models in terms of peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) metrics. Furthermore, the random forest (RF) classifier was employed for mangrove mapping. Compared to 30-m products, our 10-m map shows higher mapping accuracy and finer spatial details.**

***Index Terms*— Deep learning, high-resolution mapping, Landsat, long-term monitoring, mangroves, Sentinel-2, super-resolution.**

I. Introduction

MANGROVES are intertidal wetland ecosystems found along tropical and subtropical coastlines [1], [2]. They support a high level of biodiversity, both in terms of plant and animal species, and offer various services to humanity, including coastal protection, support for aquaculture, attraction for tourism, and climate regulation through carbon sequestration [3], [4], [5]. Nonetheless, mangroves are under threat of habitat loss due to natural disasters and human activities on a global scale [6], [7], [8]. Consequently, comprehending the distribution of mangrove species is essential for facilitating the development of mangrove conservation strategies. In recent decades, the expanding capabilities of remote sensing have led to its extensive use in mapping and monitoring the distribution of mangroves.

According to the spatial resolution of the input data, there are two main categories into which mangrove mapping studies can be classified: high spatial resolution and medium-high spatial resolution. High spatial resolution imagery (e.g., unmanned aerial vehicles, WorldView-2, SPOT, and QuickBird) provides an opportunity for accurate mangrove mapping, but such types of imagery are typically expensive and often limited to the spatial-temporal coverage, making large-scale mapping of mangroves challenging [9], [10], [11].

Currently, employing medium-high spatial resolution imagery (e.g., Landsat, Sentinel) as the input data for large-scale mangrove mapping continues to be the mainstream method [12]. Prior to 2011, our understanding of global mangrove status was primarily derived from compilations of mangrove areas in various regions or countries. These statistics were based on inconsistent data sources with varying resolutions and production methods, lacking comparability and raising doubts about their accuracy [13]. In 2011, Giri et al. produced the Mangrove Forests of the World (MFW) database, the initial globally united resolution map of mangrove forests. They employed a blend of supervised and unsupervised digital image classification methods with approximately 1000 Landsat images [14]. Building upon this, in 2016, Hansen and DeFries integrated MFW with the global forest change (GFC) database and the terrestrial ecosystems of the world (TEOW) database [15]. Through an analysis of mangrove changes, they produced the continuous global mangrove forest cover database for the 21st century (CGMFC-21). However, this method relies on preexisting map data, leading to cumulative errors. Subsequently, employing Landsat image and L-band synthetic aperture radar (SAR) along with an extreme random forest (RF) classifier, Bunting et al. [16] generated the 2010 Global Mangrove Watch (GMW) dataset in 2018. This resulted in the first globally consistent automated method for mangrove mapping. However, it should be noted that the aforementioned datasets are derived from remote sensing

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imagery with a 30-m spatial resolution. At this level of resolution, a numerous of mixed pixels leads to three issues: 1) the finer details of narrow internal river channels and open spaces were hardly identified as they blend into the surrounding surface features; 2) small patches of mangroves community tend to be omitted; and 3) it is hard to precisely delineate the edges of mangroves. These three issues can lead to inadequate classification of mangroves, thereby affecting carbon storage estimation, vegetation index assessment, and change monitoring. For example, when narrow internal river channels are merged, the substantial carbon stored within them may also be omitted [17], [18]. As a result, the existing 30-m spatial resolution imagery may not meet the increasing demand for accurate identification and monitoring analysis [19], [20], [21].

To address the challenges faced by 30-m Landsat imagery in accurately mapping mangroves, several studies have turned to using 10-m Sentinel imagery instead. For example, Zanaga et al. [22] created a worldwide land cover map for 2020 at 10-m spatial resolution (ESA World Cover 10 m v100) using Sentinel-1 and Sentinel-2 imagery. Recently, Jia et al. established a High-Resolution Global Mangrove Forests (HGMF_2020) dataset at the 10-m spatial resolution based on Sentinel-2 imagery in 2023 [23], utilizing object-based image analysis and RF classification techniques. Although these remote sensing datasets have significantly improved aspects such as edge and small patch detection compared to 30-m resolution products, these studies all depend on 10-m Sentinel imagery. However, as a result of the earliest availability of Sentinel imagery in 2014 [24], this restricts research into 10-m resolution mangrove mapping for longer time series or earlier periods. Over the past 50 years, global mangrove forests have experienced a sharp decline, with approximately 20% to 35% loss [25]. To better understand the fundamental biology and ecology of mangrove ecosystems, there is an increasing need for monitoring and assessing the dynamic changes in mangroves. This will also help guide conservation and restoration efforts and improve ecosystem management [25], [26]. To efficiently study the distribution of mangroves and monitor their temporal changes, it is promising to explore super-resolution techniques. These techniques theoretically can enhance 30-m Landsat imagery to a 10-m resolution and subsequently generate time-series mangrove products, which has not been previously attempted.

Super-resolution technology seeks to augment the spatial resolution of imagery through image processing algorithms [27], [28], [29]. In the remote sensing domain, single-image super-resolution (SISR) is typically used for reconstructing imagery with higher resolution [30], [31], [32]. However, SISR faces an inherent problem, wherein a onefold low-resolution (LR) image can have multiple super-resolution solutions rather than a unique solution. This issue can be alleviated by introducing reference images. To better leverage reference (Ref) image information, cutting-edge approaches often employ deep learning strategies based on reference images, particularly using deep convolutional neural networks to learn the mapping between corresponding LR and Ref images. Compared to nondeep learning algorithms, they have made significant advancements [33], [34], [35].

In the field of remote sensing image classification, some studies have utilized LR images in conjunction with high-resolution products for training to directly obtain high-resolution products [36], [37]. Although such methods are straightforward and do not depend on high-resolution data, they are unable to capture detailed texture information from high-resolution data, which can effectively aid in classification [38], [39], [40]. In this regard, the texture transformer network (TTSR) offers a beneficial approach. Yang et al. [41] proposed TTSR for image resolution reconstruction, which emphasizes collaborative feature learning between LR and Ref images, making the most of the information from the reference image. TTSR works in a "texture feature transfer-cross-scale feature integration" fashion. During the texture feature transfer process, the model achieves accurate texture feature transfer by calculating the texture correlation between LR and REF images, addressing issues such as texture blurring and inaccurate texture transfer. In the process of cross-scale feature integration, the model employs a cross-scale feature integration module to learn texture features at different scales (e.g., from $1\times$ to $4\times$), aiming to achieve a more powerful feature representation. While the TTSR excels in texture reconstruction, it treats features from each channel equally, which may not favor the network's representation. Residual channel attention networks (RCAN), introduced by Zhang et al., utilize a channel attention mechanism. It selectively and weightily combines channel features, performs multichannel feature fusion, and considers interdependencies among channels, enabling adaptive readjustment of intelligent channel features [42]. Nevertheless, RCAN may not perform as well in terms of texture details, with many details appearing relatively blurry. Moreover, both TTSR and RCAN have been extensively validated in the field of super-resolution image reconstruction, proving their effectiveness and outstanding performance [43], [44], [45].

The objectives of this work are threefold as follows:

1) Develop a new super-resolution model that incorporates RCAN and TTSR, namely RCAN-TTSR. This model is trained using 30-m Landsat-8 and 10-m Sentinel-2 data from the year 2020 and can generate 10-m Landsat-5 data for the years 2007–2010.

2) Create a 10-m mangrove mapping methodology by combining data from the super-resolution model with the RF method.

3) Compare and evaluate the mangrove maps derived from this study with other contemporaneous mangrove products. As such, this model can significantly improve the mapping accuracy of mangroves, especially in addressing the aforementioned issues related to narrow river channels, small patches, and boundaries. Furthermore, this study could potentially serve as a methodological reference for future larger-scale or time-series mapping of mangroves at a 10-m resolution. Additionally, the proposed model also can extend its application to diverse land cover categories for long-term mapping and analysis [46].

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II. Study Site and Data

A. Study Area

Australia boasts vast mangrove distribution, including areas like Wolcott Bay in the northwest, providing rich data resources due to its unique types and species of mangroves. Bangladesh has an extensive distribution of mangroves, with the Sundarbans being one of the world's largest mangrove conservation areas. It is renowned for its magnificent mangrove landscapes and rich biodiversity. Cuba is surrounded by water and has a profusion of mangroves. The northern mangrove wetlands are important habitats for numerous fish species, and the seagrasses and intertidal zones that fish feed on are often coexisting and supported by the coastal ecosystem's mangroves. Madagascar, located in the Indian Ocean, holds abundant mangrove resources, constituting 2% of the world's mangroves [47]. Research in this region is relatively limited, and the selected study area is in the northwest of Madagascar. Brazil has 7% of the world's mangrove forests, distributed along its vast coastline. Unfortunately, in recent years, human actions like urban expansion and aquaculture have resulted in the loss of over 50 000 hectares of mangrove forests [48]. The study area is shown in Fig. 1.

B. Datasets

1) Satellite Data: For this research, we selected three different satellite image datasets: Sentinel-2, Landsat-8, and Landsat-5.

Sentinel-2 was launched in 2015 with a spatial resolution of 10 m. For this study, we chose spectral bands 2, 3, 4, 8, 11, and 12 of this imagery, covering the entire year of 2020 [49], [50].

Landsat-8 was launched by NASA in 2013, providing a spatial resolution of 30 m. We selected spectral bands 2, 3, 4, 5, 6, and 7 for this study, also spanning the entire year of 2020 [51].

Landsat-5, launched by NASA in 1984 and decommissioned in 2013, had a spatial resolution of 30-m. For this research, we used the spectral bands 1, 2, 3, 4, 5, and 7 of Landsat-5, covering the time frame from 2007 to 2010 [52].

All Landsat and Sentinel data were acquired as surface reflectance (SR) data and processed and downloaded using the Google Earth Engine (GEE). The platform allows users to directly access preprocessed (e.g., atmospheric correction and radiometric calibration) SR data, avoiding the cumbersome process of large-scale image preprocessing.

2) Reference Data: To acquire sufficient reference data for the RF classifier, including training and validation samples, we visually interpreted Landsat-5 images after resolution enhancement on GEE. A total of 2950 samples were gathered within the study area, distributed across regions as follows: 586 mangrove and 610 nonmangrove samples in Bangladesh, 209 mangrove and 201 nonmangrove samples in Cuba, 192 mangrove and 232 nonmangrove samples in Madagascar, 273 mangrove and 228 nonmangrove samples in Australia, and 207 mangrove and 212 nonmangrove samples in Brazil. Due to varying region sizes, sample numbers differed among regions, with efforts made to maintain a balanced ratio between mangrove and nonmangrove samples and ensure a relatively even distribution across the study area. Finally, all samples were randomly split into training and validation sets at a 4:1 ratio.

III. Methods

This study employed Landsat-8 and Sentinel-2 data as training data to reconstruct the resolution of Landsat-5 imagery. The approach consisted of three parts. First, perform a series of preprocessing on the study area images. Second, employ a deep learning-based super-resolution model for resolution enhancement and evaluate the reconstruction results. In the end, the mapping of mangroves was conducted using the RF method, and the classification results were evaluated. The flowchart proposed in this study is shown in Fig. 2.

A. Data Preprocessing

Before training data with the super-resolution model, the data needs to be processed, which involves two steps as follows:

1) To reduce the impact of cloud cover on image quality before training the study area images, cloud masking is performed on the GEE platform. Specifically, for Sentinel-2 imagery, we utilize the "QA60" band with cloud mask information [53], masking out the opaque cloud and cirrus cloud pixels flagged in bits 10 and 11 of the "QA60" band. For Landsat-8 and Landsat-5 imagery, we use the "pixel_qa" band containing cloud mask information, masking the opaque cloud and cirrus cloud pixels flagged in bits 3 and 5 of the "pixel_qa" band. Additionally, to ensure the quality of the required images, we employed a median synthesis method, synthesizing Sentinel-2 imagery into composite images for the entire year of 2020, and likewise, Landsat-8 imagery was synthesized into composite images for the same period. Similarly, Landsat-5 imagery from 2007 to 2010 was synthesized into a single composite image. Median compositing is applied to multiple images of the same study area, which is less susceptible to potential extreme outliers [54]. The Landsat and Sentinel SR data we used on the GEE platform have undergone atmospheric correction and radiometric calibration. In addition, we transformed Landsat-5, Landsat-8, and Sentinel-2 images to the EPSG:4326 coordinate system and downloaded them to the same extent, ensuring spatial alignment and similar textures. Upon obtaining the images, due to the pixels of Landsat-8 and Sentinel-2 images being not completely aligned at the image boundary, the ratio of pixels between the two images does not reach 1:3, making training infeasible. To address this issue, aligning the two images with different resolutions at the pixel level is required, ensuring that each Landsat-8 pixel comprises nine Sentinel-2 pixels, as depicted in Fig. 3. Specifically, the procedure involves initially using a vector as a mask for extracting the Landsat-8 image. Then, the Landsat-8 image is used as a mask to extract the Sentinel-2 image. This operation can be performed using the "extract by mask" feature in ARCGIS.

2) To train the model, we extracted patches of size 256 × 256 pixels from each Landsat-8 image along with corresponding patches of size 768 × 768 pixels for the Sentinel-2 image.

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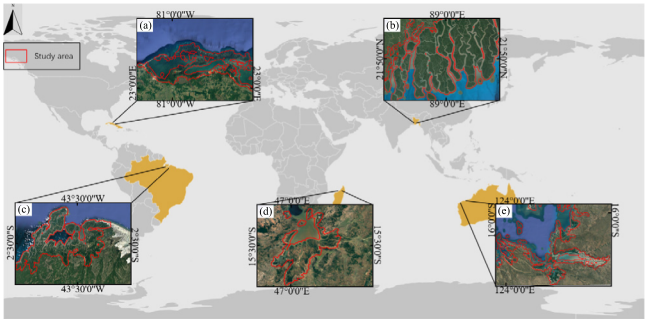


Fig. 1. Site of the study area. (a) Cuba. (b) Bangladesh. (c) Brazil. (d) Madagascar. (e) Australia.

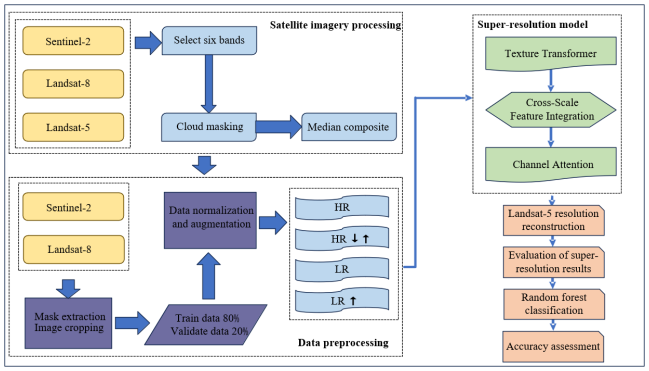


Fig. 2. Flowchart of the proposed method.

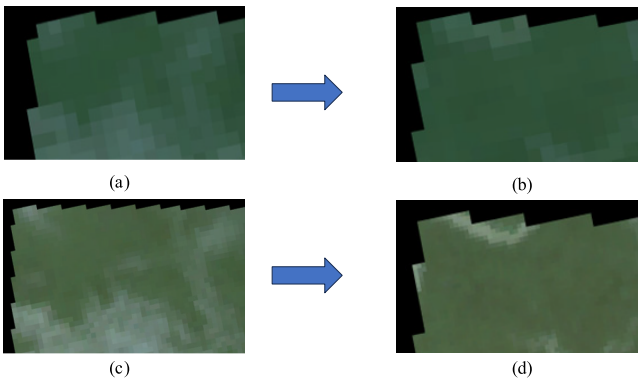


Fig. 3. Imagery extraction is shown in the following figure. (a) Landsat-8 imagery before extraction. (b) Landsat-8 imagery after extraction. (c) Sentinel-2 imagery before extraction. (d) Sentinel-2 imagery after extraction.

To ensure that images from different sensors were within the same value range, we normalized the image values to (0,1).

Additionally, to accurately capture the texture features of the Sentinel-2 imagery for our model to learn, we utilize not only the Sentinel-2 and Landsat-8 images as inputs but also performed up-sampling and down-sampling on the Sentinel-2 images (Sentinel-2↑↓), along with up-sampling on the Landsat-8 imagery (Landsat-8↑). After up-sampling, Landsat-8↑ attains the same spatial size as Sentinel-2. It establishes a correlation between Sentinel-2 and Landsat-8 images, enabling the assessment of areas of similarity or dissimilarity. In regions of similarity, the texture from Sentinel-2 images is transferred, while dissimilar textures are concurrently suppressed. Down-sampling and up-sampling Sentinel-2 images result in a blurred Sentinel-2 image (Sentinel-2↓↑)

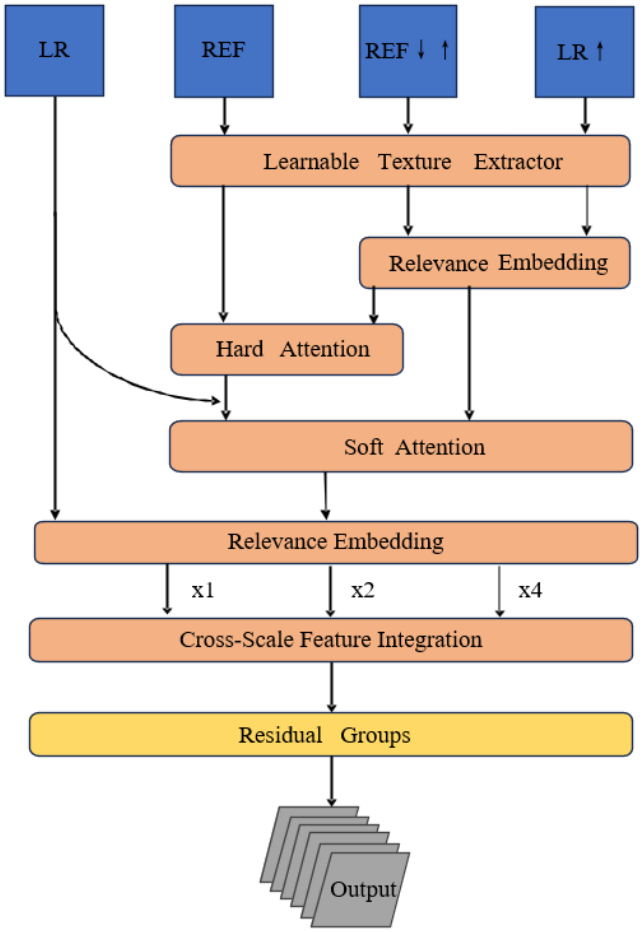


Fig. 4. Structure of the RCAN-TTSR model.

that matches the frequency band of Landsat-8↑. With similar clarity, the assessment of texture similarity becomes more accurate, leading to more precise identification of texture-similar locations. Consequently, the model utilizes Landsat-8, Landsat-8↑, Sentinel-2, and Sentinel-2↑↓ as its input data. About 80% of the data from these four input datasets were selected for model training, while the remaining 20% were utilized for model validation.

B. Network Architecture of the RCAN-TTSR

This study proposes a model based on RCAN and TTSR, which includes three parts: texture transformer, cross-scale feature integration, and channel attention module. As shown in Fig. 4, the texture transformer determines the texture features to be transferred from HR, which assists in obtaining better texture results. The cross-scale feature integration fuses the image information of different resolutions to improve the quality of the generated images, and the channel attention module adaptively adjusts the weights of the channels to improve the network's ability to improve network performance.

1) Texture Transformer: The texture transformer consists of four parts: learnable texture extractor, relevance embedding, and hard attention and soft attention.

The feature extractor takes both Sentinel-2 and Landsat-8 as inputs, as well as Sentinel-2↑↓ and Landsat-8↑. This design enables Sentinel-2 and Landsat-8 imagery to be in the same

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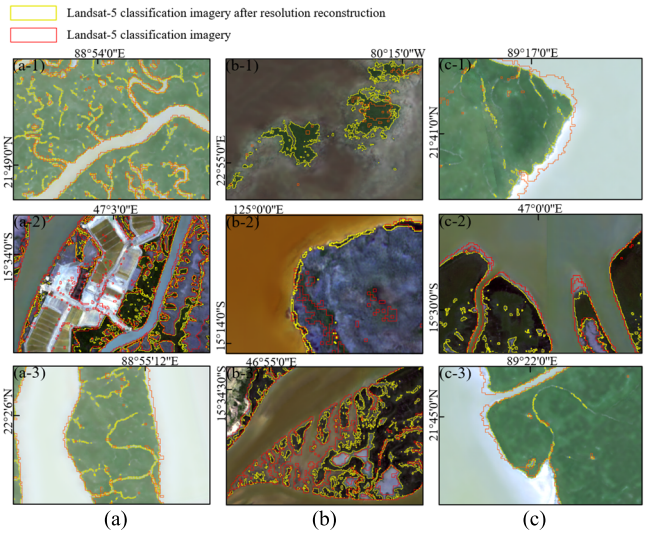


Fig. 5. Comparison between Landsat-5 classification imagery after resolution reconstruction and Landsat-5 classification imagery in terms of the finer details of narrow internal river channels and open spaces, small patches of mangroves, and the edges of mangroves. The base map consists of Landsat-5 imagery after resolution reconstruction with RGB bands. (a) Creeks and open spaces of mangroves. (b) Small patches of mangroves. (c) Edges of mangroves.

domain, improving the accuracy of feature comparison and assisting in obtaining better texture results.

The Relevance Embedding module evaluates the correlation between Sentinel-2↑↓ and Landsat-8↑ for texture transfer by outputting a Hard Attention map and a Soft Attention map. To be more precise, the module unfolds both images into feature blocks and then calculates the correlation between feature blocks by computing the normalized inner product. The larger the normalized inner product, the stronger the correlation between the two feature blocks, and the more high-frequency texture information that can be transferred.

The Hard Attention module transfers Sentinel-2 texture features extracted from the Sentinel-2 image. The hard attention map records the location of the most relevant feature block in Sentinel-2↑↓ for each feature block in Landsat-8↑. Afterward, the Hard Attention module utilizes these recorded positions to transfer the corresponding feature blocks from Sentinel-2 texture features and combines them into a transferred texture feature map.

In the soft attention module, the transferred texture feature map is first fused with the Landsat-8 feature to take advantage of more information from the Landsat-8 image, and then the soft attention map is multiplied with the fused feature to enhance the transmission of relevant textures and suppress the transmission of irrelevant textures.

2) Cross-Scale Feature Integration: In the cross-scale feature integration module, the texture transformer outputs at the three resolution scales (×1, ×2, and ×4) are cross-fused, and the feature channels are then mapped to the original channel number. This allows image information at different resolution scales to be fused into the generated image, resulting in a more powerful feature representation and improved quality of the generated images.

3) Channel Attention: The introduction of a channel attention mechanism is necessary because different channels contain unique feature information. Therefore, the network needs to focus on the most useful information for resolution reconstruction by using residual channel attention block (RCAB). First, global average pooling is used to obtain the weights of each channel. Second, an activation function combines the weights with the input channel feature information, making the network generate different attention for each channel feature.

4) Loss Function: In image super-resolution tasks, the loss function plays a crucial role in measuring the difference between the generated image and the target image. For this study, the L1 loss function is employed for training and evaluation because of straightforward implementation and excellent performance in experiments. The L1 loss function is one of the most commonly used pixel-level difference measurement functions. Moreover, compared with the L2 loss function, it can accelerate the convergence speed and produce higher objective evaluation scores. The expression is as follows:

$$\text{Loss1} = \frac{1}{m}\sum_{i=1}^{m} |y_i - f(x_i)| \tag{1}$$

where y_i represents Sentinel-2 imagery and $f(x_i)$ represents the Landsat imagery after resolution reconstruction.

5) Model Training and Parameter Settings: During the training process, the initial learning rate, batch size, and number of epochs were configured as follows: 0.001, 64, 50. For optimization, we employed the Adam optimizer for loss function optimization: b1 = 0.9, b2 = 0.999, and $e = 10^{-7}$. Since we need to train six bands of remote sensing imagery, the channel numbers of the first and last Conv layers in our model were set to 6. In addition, 80% of the data was used as training data, and 20% was used as validation data. The number of channel attention blocks in the model was set to 20, and the variance and mean used were calculated from the selected study area. Peak signal-to-noise ratio (PSNR) was used as the metric for evaluating the trained model, and the best-performing model in terms of PSNR was selected as the optimal prediction model. The most effective predictive model was employed to reconstruct the resolution of Landsat-5 imagery. The experiments took place on a computer running Windows 10 equipped with an RTX 3090 GPU and 24 GB of memory, using the PyTorch framework for deep learning training and prediction.

C. Evaluation of Super-Resolution Results

To better evaluate the effectiveness of resolution reconstruction, we use commonly used PSNR and structural similarity index (SSIM) as fidelity metrics to measure the difference between the super-resolution image and the high-resolution image. Given a real image x and its corresponding super-resolution image y, the PSNR and SSIM can be calculated to evaluate the experimental results. The higher the PSNR and SSIM values, the more consistent the generated image is with the reference image. The formula for calculating PSNR is as follows:

$$\text{PSNR}(x, y) = 10\log_{10}\frac{L^2}{\text{MSE}(x, y)} \tag{2}$$

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TABLE I
COMPARISON OF PSNR/SSIM AMONG FOUR SR METHODS FOR DIFFERENT REGIONS

	RCAN-TTSR	RCAN	TTSR	SRGAN	SWINIR	BICUBIC
Bangladesh	**40.86**/0.9685	39.32/0.9633	39.76/0.9635	38.09/0.9520	37.70/0.9429	33.41/**0.9792**
Cuba	**38.13**/**0.9657**	36.73/0.9557	37.92/0.9633	33.13/0.8951	32.63/0.8780	31.55/0.6789
Madagascar	**39.86**/**0.9636**	38.29/0.9493	39.56/0.9589	36.68/0.9197	36.17/0.9119	33.60/0.8864
Brazil	**38.69**/0.9530	35.83/**0.9833**	38.50/0.9527	35.46/0.9027	34.81/0.8916	34.40/0.8638
Australia	**41.59**/**0.9679**	39.02/0.9601	40.77/0.9633	40.31/0.9046	40.76/0.9077	33.42/0.8988

where L is a constant representing the maximum dynamic range of the image data type, but since we have normalized the images, we set L to 1. MSE (x, y) represents the mean square error between x and y. The formula for calculating SSIM is as follows:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{3}$$

where μ_x represents the mean, σ_x represents the standard deviation, and σ_{xy} represents the covariance.

D. Mangrove Mapping With RF

After performing resolution reconstruction, we utilized the RF method in GEE to classify the mangroves from the Landsat-5 imagery that had undergone resolution reconstruction. The RF method is one of the most popular classification algorithms in remote sensing research due to its speed and ease of use. The RF method generates predicted class outputs by making a large number of decision trees in the time of the training process [55]. The classification of tree species has been shown to be effective using the RF classifier in several studies [56], [57]. We set the number of trees to 200, as it has been proven to yield good results. The detailed code can be viewed at the https://code.earthengine.google.com/181d977156d933550fc1348cc96b8f19 and https://code.earthengine.google.com/e7db358c5232e661b3c7fae2bab9a15d.

E. Accuracy Assessment

To fully demonstrate the accuracy of the 10 m mangrove products generated by the RCAN-TTSR model, this study compared it with two other products produced during the same period: 1) compared with Landsat-5 images and 2) compared with the GMW v3.0 product.

For the first comparison, we used the RF method to classify the mangroves from Landsat-5 images. Evaluating the classification accuracy of the image prior to resolution reconstruction is beneficial for visually assessing the effectiveness of resolution reconstruction.

For the second comparison, GMW v3.0 is a global mosaic dataset consisting of 11 time periods produced by the Japan Aerospace Exploration Agency (JAXA) using the L-band SAR from 1996 to 2020. We selected the time of 2010 in this research.

We evaluated the results using two different indicators for the above two products: overall accuracy (OA) and Kappa coefficient. The OA quantifies the proportion of reference pixels mapped correctly by our mangrove classification method. The Kappa coefficient is often used to test the spatial consistency of image classification, and the larger the Kappa coefficient, the better the classification effect.

IV. RESULTS

A. Quality Evaluation of Super-Resolution Results

To comprehensively showcase the precision of the RCAN-TTSR approach, we compared our proposed method with other super-resolution methods in five study areas (as shown in Table I). The compared SR methods include four methods [RCAN-TTSR, RCAN, TTSR, super-resolution generative adversarial network (SRGAN), image restoration using swin transformer (SWINIR), and bicubic interpolation (BICUIC)]. All these methods were fully trained on our study area data to obtain the best performance for fair comparison. We used the PSNR and SSIM metrics to quantitatively evaluate the SR results. In each row, the most outstanding outcome in each row is denoted in bold font.

For PSNR, RCAN-TTSR outperformed other super-resolution methods in all study areas, with improvements of 1.1, 0.21, 0.34, 0.19, and 0.82 dB over the second-best method in each study area.

For SSIM, RCAN-TTSR performed better than other super-resolution methods in Cuba, South Africa, and Australia, with improvements of 0.0024, 0.0047, and 0.0046 dB over the second-best method in each study area, respectively. For the Bangladesh study area, the best-performing method is BICUBIC, with an improvement of 0.0107 dB over the second-best method. For images with complex textures, deep learning-based super-resolution methods typically outperform BICUBIC interpolation in preserving details and enhancing image quality. However, in the case of the extensive and continuous mangrove areas in the Bangladesh region, there is a relative lack of complex textures or high-frequency details compared to other regions. This provides an opportunity for BICUBIC to perform well in the SSIM metric, surpassing the scores of RCAN-TTSR and emerging as the top-performing super-resolution method in terms of SSIM in the Bangladesh region. For the Brazil study area, the best-performing method is RCAN, with an improvement of 0.0303 dB over the second-best method. In comparison to other regions, the Brazil region has a higher proportion of urban areas. Nevertheless, the training data used for RCAN-TTSR contains fewer images related to urban areas. This leads to inaccurate texture transfer

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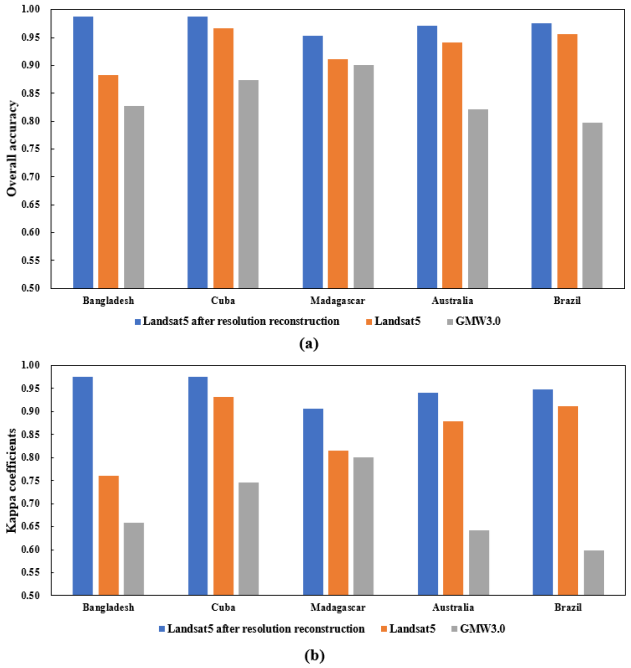


Fig. 6. (a) OA of resolution-reconstructed images, Landsat-5 images, and GMW v3.0 in the study area. (b) Kappa coefficients of resolution-reconstructed images, Landsat-5 images, and GMW v3.0 in the study area.

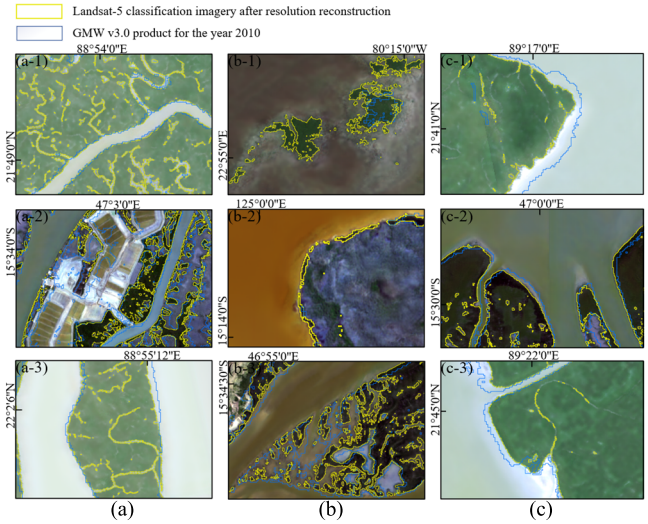


Fig. 7. Comparison between Landsat-5 classification imagery after resolution reconstruction and GMW v3.0 product for the year 2010 in terms of the finer details of narrow internal river channels and open spaces, small patches of mangroves, and the edges of mangroves. The base map consists of Landsat-5 imagery after resolution reconstruction with RGB bands. (a) Creeks and open spaces of mangroves. (b) Small patches of mangroves. (c) Edges of mangroves.

by RCAN-TTSR into the results for urban regions, resulting in lower SSIM scores.

Overall, our method performed well in most cases. In addition, we noticed that the traditional BICUBIC method exhibits a significant performance gap in evaluation metrics compared to other super-resolution methods in most regions. The reason may be that this method only considers the grayscale values of adjacent points and cannot generate new details.

B. Comparison With Other Products

We compared the accuracy and visual effect of the obtained products with two other products: Landsat-5 imagery and the GMWv3.0 product. In response to the three limitations identified in the introduction regarding the 30-m spatial resolution, namely the challenges in identifying the finer details of narrow internal river channels and open spaces, the tendency to omit small patches of mangroves, and the difficulty in precisely delineating the edges of mangroves, significant improvements in classification results have been achieved through resolution reconstruction.

For visual comparison, we have selected three typical areas for each aspect: the creeks and open spaces, small patches, and edges of mangroves, as shown in Fig. 5. The Landsat-5 classification imagery after resolution reconstruction can identify numerous narrow internal river channels and accurately distinguish open spaces from mangroves, as shown in Fig. 5(a-1) and (a-3). In contrast, the original Landsat-5 classification imagery can only partially identify narrow internal river channels and often misclassifies open spaces and buildings, as shown in Fig. 5(a-2). Regarding small patches of mangroves, the Landsat-5 classification imagery after resolution reconstruction not only identifies mangrove patches that are omitted in the original Landsat-5 classification imagery, as shown in Fig. 5(b-1) and (b-2), but also separates merged patches into individual ones, as shown in Fig. 5(b-3). For the edges of mangroves, it is observed that the Landsat-5 classification imagery after resolution reconstruction aligns better with the real mangrove edges compared to the original Landsat-5 classification imagery, as shown in Fig. 5(c-1)–(c-3).

1) Comparison With Landsat -5 Imagery: According to the classification accuracy shown in Fig. 6. For five study areas, the OA of mangrove classification products using Landsat-5 is 88.3%, 96.6%, 91.1%, 94.1%, and 95.6%, and the Kappa coefficients were 0.76, 0.93, 0.82, 0.88, and 0.91, respectively. In contrast, the OA of the classification products extracted from super-resolution imagery is 98.8%, 98.8%, 95.3%, 97.1%, and 97.5%, with corresponding Kappa coefficients of 0.98, 0.98, 0.95, 0.94, and 0.95.

The visual effects are shown in Fig. 7. In terms of the finer details of narrow internal river channels and open spaces, the GMW v3.0 product faces challenges compared to the mangrove product based on Landsat-5 imagery after resolution reconstruction. This is mainly because many narrow internal river channels are misclassified as mangrove forests in GMWv3.0 product, leading to their omission, as shown in Fig. 7(a-1) and (a-3). Additionally, some open spaces and buildings are incorrectly classified as mangroves, as shown in Fig. 7(a-2). Regarding small patches of mangroves, the GMW v3.0 product omits or merges these mangrove patches, which are accurately identified by the mangrove product based on Landsat-5 imagery after resolution reconstruction, as shown in Fig. 7(b-1)–(b-3). For the edges of mangroves, it can be observed that GMW v3.0 exhibits significant jaggedness and erroneously classifies water bodies outside mangroves as mangroves, resulting in poor alignment with the mangrove edges

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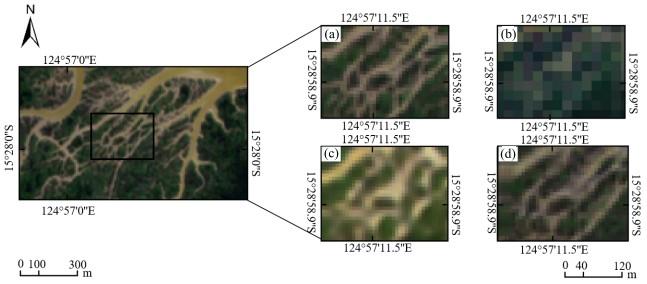


Fig. 8. (a) Sentinel-2 image. (b) Landsat-8 image. (c) Before texture feature transfer. (d) After texture feature transfer.

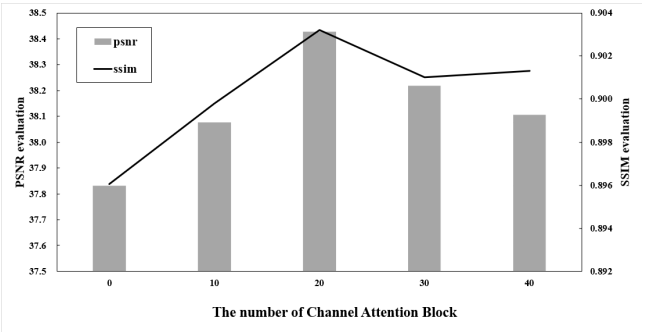


Fig. 9. The model's performance was assessed with varying numbers of channel attention blocks, and the average results across the five study areas were obtained.

and consequently leading to an overestimation of mangrove area, as shown in Fig. 7(c-1)–(c-3).

2) Comparison With the GMW v3.0 Product: For the classification accuracy, as depicted in Fig. 6, GMW v3.0 products show OA in the five study areas of 82.7%, 87.4%, 90.1%, 82.1%, and 79.8%, with matching Kappa coefficients of 0.66, 0.75, 0.8, 0.64, and 0.6, respectively. In contrast, the classification accuracy of the products extracted from the super-resolved images was improved. The overall accuracies in all study areas exceeded 95%, and the matching Kappa coefficients were all above 0.9. The significant differences between the two products will be further discussed in Section V-B.

V. Discussion

A. RCAN-TTSR

This article introduces a deep learning-based super-resolution model (RCAN-TTSR) that outperforms five other super-resolution methods in most study areas. This is mainly attributed to two factors: 1) RCAN-TTSR effectively learns the texture features of Sentinel-2 imagery and 2) RCAN-TTSR reweights the importance of different channels.

1) Texture Feature Transfer: RCAN-TTSR effectively learns the texture features of Sentinel-2 imagery. Prior studies indicate that abundant textures from high-resolution images can compensate for details lost in corresponding LR images [58], [59], [60]. Specifically, it involves transferring high-resolution textures to LR images to yield visually pleasing outcomes. For instance, Zheng et al. [61] employed a U-Net structure for multiscale fusion of high and LR images. However, this direct method of transferring textures may lead to inaccuracies in search and transfer. In this study, the texture transformer used by RCAN-TTSR calculates the correlation between Landsat-8 and Sentinel-2 images through inner products. Additionally, it assigns weights to texture information, effectively searching for and transferring high-resolution texture features to LR images.

We selected representative regions and compared texture features before and after transfer to assess the effectiveness of texture feature transfer.

As shown in Fig. 8, through visual comparison of the mangrove imagery before and after texture feature transfer, we found that the imagery after texture feature transfer better restored the texture details of the Sentinel-2 image. In addition, the mangrove image after texture feature transfer can clearly show the narrower internal river channels inside the mangrove compared to before texture feature transfer, and identify the mangrove patches that are easily omitted.

Despite the advantages of RCAN-TTSR in texture feature transfer, there are still limitations that need to be noted. Texture feature transfer requires high-resolution images to have similar content to the LR images. Due to the persistent cloud cover in tropical and subtropical regions, the number of available images is severely reduced. We selected the time range for Sentinel-2 and Landsat-8 images on the GEE platform for the entire year of 2020, which means that the content of Sentinel-2 and Landsat-8 images may not be completely identical, which may affect the effect of texture transfer.

2) Channel Attention Block: RCAN-TTSR reweights the importance of different channels. Specifically, it flexibly adjusts the output by using the feature information from different channels to obtain different weights, select useful information for reconstruction, and discard irrelevant feature information. Previously, Hu et al. [62] improved the network's expressive ability by modeling the interchannel dependency of convolutional features and adaptively recalibrating channel feature responses. The channel attention block applies this idea to image super-resolution, enhancing the model's performance by increasing the weight of important channels and suppressing the weight of less important channels.

To gauge the channel attention block's efficacy, we compared models with different numbers of channel attention blocks in the five study areas, using PSNR and SSIM as evaluation metrics. The results were the average values of the five study areas.

In Fig. 9, we show the PSNR and SSIM obtained by models with varying numbers of 0, 10, 20, 30, and 40 Channel Attention Blocks. The network with channel attention blocks achieved the highest performance in both evaluation metrics when the number of channel attention blocks was 20.

Despite the advantage of RCAN-TTSR in channel attention, there are still limitations that need to be noted. The super-resolution model is developed for RGB images with three bands, but we applied it to remote sensing images with six bands, which may have unpredictable effects on the performance of the model.

B. Analysis of the Super-Resolution Mangrove Product

This article presents a novel deep learning-based super-resolution model (RCAN-TTSR) trained on images from

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2020 to generate high-resolution images from 2010. Subsequently, the high-resolution images were used to create a mangrove map using RF. Compared to other mangrove products from the same period, the super-resolution mangrove product showed better visual effects and accuracy evaluation, which is supported by four aspects.

First, the super-resolution mangrove product has a higher spatial resolution than other products from the same period. A previous study reported that a 10-m resolution image is helpful for discriminating small mangrove patches [19]. The higher resolution of the super-resolution mangrove product allows for better identification of small mangrove patches, narrower internal river channels, and open spaces, leading to superior OA and Kappa coefficients compared to other mangrove products from the same period.

Second, the RF classification method has excellent classification results [63]. By comparison, GMW v3.0 uses L-band SAR for detecting mangrove changes. Although L-band radar data are sensitive to mangrove changes, its classification ability for mangrove extents is limited [64], [65]. For example, this product misclassifies narrower internal river channels between mangroves as mangroves, leading to significant differences in OA and Kappa coefficients compared to other products, especially in the Australian and Brazilian regions.

Third, the super-resolution mangrove product does not require assistance from other products. In contrast, GMW v3.0 utilizes data from sources like SRTM to refine mangrove boundaries and delineate mangrove habitat layers, minimizing unnecessary processing significantly. This approach proves highly effective for global-scale mangrove mapping. However, there are also some limitations to using additional data sources in GMW v3.0, especially when GMW v3.0 extracts mangrove samples and nonsamples from other mangrove products such as Giri et al. [14] and Spalding [66] for training. First, it may lead to error accumulation, particularly when the accuracy and reliability of these data sources cannot be guaranteed. Second, differences in spatial resolution and temporal coverage among different data sources may affect the accuracy of the final mangrove mapping results.

Despite the advantages in visual effects and classification accuracy, there are still limitations that need to be noted in the super-resolution mangrove product. This study utilized three different types of data (Landsat-8, Landsat-5, and Sentinel-2) to train and predict RCAN-TTSR. However, because each has a different radiance response, this theoretically could affect the model's applicability. Fortunately, these three data types' six bands (Red, Blue, Green, near infrared (NIR), SWIR1, and SWIR2) exhibit significant linear correlations. Deep learning is adept at handling such linear correlations, so this study disregarded the impact of radiance response differences on the resolution reconstruction. In future research, improving the proposed model can be achieved by incorporating the relationships among these three data types. The relationship between Landsat-8 and Sentinel-2 is illustrated in Appendix A, with the coefficient of determination (R^2) values ranging from 0.70 to 0.96 for the six different bands. Existing studies have also shown that Landsat-8 and Landsat-5 have consistent radiometric responses [67], [68]. Additionally, the training data of the super-resolution mangrove product were processed using median compositing in GEE to eliminate the extremes caused by clouds and shadows, but the median compositing method may not highlight the peak values of the data, which could be disadvantageous to the features of the image.

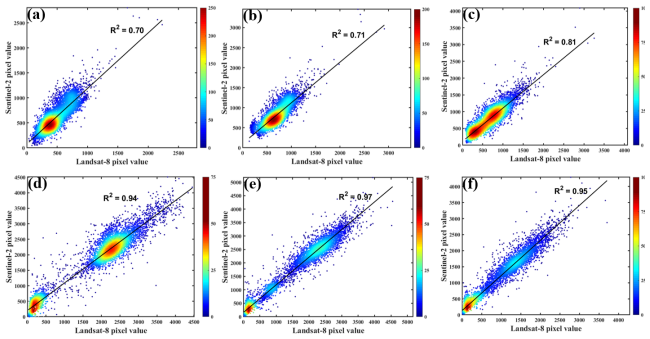


Fig. 10. Radiometric response of Landsat-8 and Sentinel-2. (a) Red band, (b) Green band, (c) Blue band, (d) NIR band, (e) SWIR1 band, and (f) SWIR2 band. We have provided a total of 9000 sample points across various regions, allocated based on the proportional area of each region. Specifically, Bangladesh has 1200 samples, Madagascar has 730 samples, Brazil has 570 samples, Australia has 5000 samples, and Cuba has 1500 samples.

VI. Conclusion

This article presents a novel deep learning-based super-resolution model and combines it with mangrove mapping using RF for long-term 10-m resolution mangrove mapping. This method not only allows LR images (such as Landsat-8 and Landsat-5) to learn high-resolution texture features (such as Sentinel-2), but also reweights the importance of different channels. In addition, we found that using a 10-m resolution significantly improves mangrove mapping accuracy, especially for resolving issues related to narrow internal river channels, small patches, open spaces, and boundaries. In conclusion, this research can potentially be a valuable methodological reference for future larger-scale and time-series mapping of mangroves at a 10-m resolution. Furthermore, it may be applied to other land cover types for 10-m long-term mapping with Landsat image.

Appendix A
The Relationship Between Landsat-8 and Sentinel-2

See Fig. 10.

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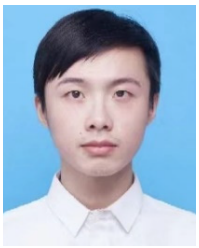
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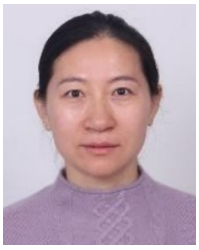
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